

AI-Powered Customer Churn Analysis

1. Introduction

Customer churn refers to customers leaving a company's services. High churn negatively affects revenue and customer growth. This project aims to identify factors contributing to customer churn and provide actionable business insights.

2. Business Problem

Telecommunication companies lose significant revenue when customers discontinue their subscriptions.

Objective:

- Identify customer groups with high churn risk.
- Understand factors affecting customer retention.
- Provide business recommendations to reduce churn.

3. Dataset Overview

Dataset Name: Telco Customer Churn Dataset

Dataset Information

Metric	Value
Total Customers	7043
Total Features	21
Churn Customers	1869
Non-Churn Customers	5174

Important Variables

- Gender
- Senior Citizen
- Partner
- Dependents

- Tenure
- Internet Service
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges
- Churn

4. Tools and Technologies Used

Python

Used for:

- Data Cleaning
- Exploratory Data Analysis
- Machine Learning

Excel

Used for:

- Data Validation
- Pivot Analysis

PostgreSQL

Used for:

- Business Queries
- Customer Segmentation

Tableau Public

Used for:

- Interactive Dashboard Creation

5. Data Cleaning Process

The following cleaning steps were performed:

- Checked missing values.
- Converted TotalCharges column to numeric datatype.
- Verified dataset consistency.
- Removed blank values.

Final dataset size remained:

7043 rows × 21 columns.

6. Exploratory Data Analysis

Average Tenure

- Churn Customers: 17.98 months

- Non-Churn Customers: 37.57 months

Average Monthly Charges

- Churn Customers: Higher monthly charges.
- Non-Churn Customers: Lower monthly charges.

Contract Analysis

Month-to-month customers exhibited the highest churn rate.

Internet Service Analysis

Fiber optic customers showed higher churn behavior.

Payment Method Analysis

Electronic check customers experienced the highest churn.

7. SQL Analysis

SQL queries were used to:

- Count churned customers.
- Analyze contract distribution.
- Compare payment methods.
- Study internet service behavior.

8. Machine Learning Model

Model Used

Logistic Regression

Features Used

- Tenure
- Monthly Charges
- Total Charges

Model Accuracy

79.7%

Interpretation

Customers with shorter tenure and higher monthly charges are more likely to churn.

9. Tableau Dashboard

Dashboard Components:

KPI Cards

- Total Customers
- Churn Customers
- Churn Rate
- Average Monthly Charges

Charts

- Contract vs Churn
- Internet Service vs Churn
- Payment Method vs Churn
- Average Tenure by Churn

Interactive Filters

- Contract
- Internet Service
- Payment Method

10. Key Insights

1. Month-to-month contracts have the highest churn.
2. Fiber optic customers are more likely to leave.
3. Electronic check customers show higher churn behavior.
4. Customers with longer tenure are more likely to stay.

11. Business Recommendations

Recommendation 1

Encourage customers to switch from month-to-month plans to long-term contracts.

Recommendation 2

Improve customer experience for fiber optic users.

Recommendation 3

Investigate issues related to electronic payment users.

Recommendation 4

Target newly acquired customers with retention campaigns.

12. Conclusion

This project successfully combined SQL, Python, Excel, Machine Learning, and Tableau to understand customer churn behavior and generate actionable business insights that can support customer retention strategies.